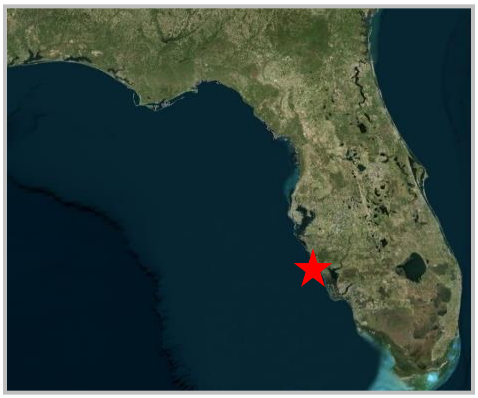
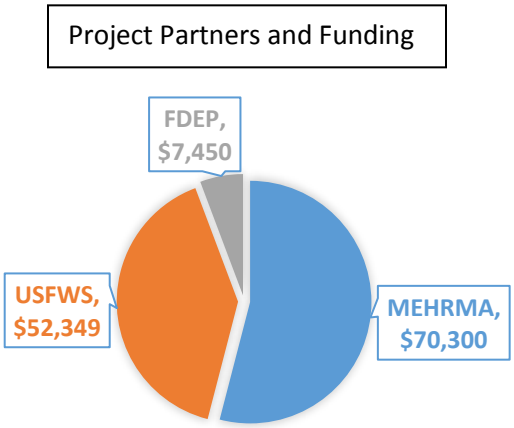
Don Pedro Island State Park Restoration of Tidal Marsh Hydrology and Habitats, Charlotte County, FL
FY 16/17





Project Description

Prior to state acquisition of Don Pedro Island State Park, dredged material from a canal adjacent to the park was placed along the south park boundary creating an artificial berm that restricted natural hydrologic flows and wildlife access to the saltmarsh and mangrove habitats. The goal of this project was to restore historic tidal connections and flows to approximately 20 acres of saltmarsh and mangrove habitats. This was achieved by the removal of 2.79 acres of the dredge spoil along the 994 ft. canal bank. Locations where dredged material is removed is being replanted with native species found in the adjacent saltmarsh and mangrove/oyster communities, which will aid in shoreline stabilization and in enhancement of the habitats. This project increased tidal exchange on a north/south continuum with the waters of Lemon Bay, a state aquatic preserve and a portion of the Charlotte Harbor National Estuary focus area. The restored tidal connection and enhanced habitats will provide a nursery and foraging areas for a variety of wildlife species, including: juvenile fish, wading birds, manatees, invertebrates, and oysters. Also, this enhanced saltmarsh habitat will benefit the state threatened golden leather fern which is present in the marsh. This project is within a state park and is part of a larger effort to restore the previously disturbed area to its natural condition and function.



Figure 1: Newly planted smooth cordgrass, needle rush and red mangroves. August 9, 2017.

The Don Pedro Island State Park project utilized $1,400 of OPS funding for aerial photography to document the site before and after construction. In addition to state sources the project received funding from the US fish and Wildlife Service through the National Coastal Wetland Conservation Grant Program. The funding provided by MEHRMA and FDEP was used as match towards the federal grant which is awarded through September 30, 2018. National Coastal Wetland

Conservation Grant funds will be used for native plantings (Figure 1), invasive vegetation treatment on 15 acres of the park, improving two miles of fire line, and feral hog trapping and removal.

Benefits

- Eliminate seed source of exotic upland plants
- Provide substrate and appropriate hydrology for mangroves and salt marsh plants
- Provide nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, and wading birds.

FWC Strategic Plan Elements Supported

Theme 1, Goal 1. Ensure the sustainability of Florida's fish and wildlife populations.
Theme 1, Goal 2. Ensure that sufficient habitat exists to support healthy populations of all native species of fish and wildlife.
Theme 3, Goal 3. Increase opportunities for residents and visitors, especially youth, to actively support and practice fish and wildlife conservation stewardship.
Theme 3, Goal 4. Encourage communities to conserve lands and waters critical to sustaining healthy and diverse populations of fish and wildlife.

Additionally, this project supports the following MESS Strategic Plan and FWRI mission goals:
• Strategic Action 1.1.1: Identify and develop restoration projects in marine and estuarine habitats (e.g. mangroves, seagrass, saltmarsh, oyster reef, coral reef) on publicly-owned state lands including those on conservation lands acquired through non-regulatory means.
• Strategic Action 1.1.4: Implement priority restoration plans.

Acreage restored by habitat type (footprint)

Total- 2.79 acres saltmarsh:

- *Spartina*, *Juncus,* and mangrove marsh- 2.79 acres

Acreage affected by restoration activities: 22.79 acres (includes acreage listed above and 20 acres of mangrove and saltmarsh.)



Figure 2: Spoil area prior to removal. East of project site viewed to the west. May 20, 2017.



Figure 3: Spoil area after berm removal and grading. East of project site viewed to the west. July 11, 2017.